Spatial Mass distributions and Galactic Stability

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Overview

- Aim
- Project Overview
- Status

Aim of my research

- Dark Matter has shown little experimental success.
- Can a universal non-constant mass distribution remedy the aperent motion?

Aim of my research

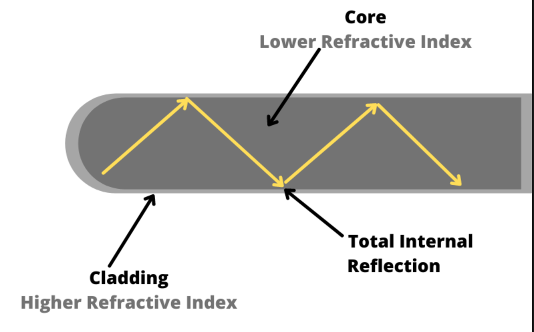

Figure: Courtesy of https://www.cspsprotocol.com/fiber-optics/

Project Overview

- Warm up: find lagrangian associated with $z(t) = \frac{\sqrt{a^2t^2+1}-1}{a}$
- Warm up: by proxy mass distribution $\mathcal{L} = -m(z)\sqrt{1-\dot{z}^2}$
- Main: Find the trajectory that desribes percieved motion in galaxies and repeat the above process.
- Main: Compare to data

Status

- I have found $m(z) = \frac{m_0}{az+1}$ and $\mathcal{L} = -\frac{m_0}{az+1}\sqrt{1-\dot{z}^2}$
- I have also proven $z(t) = \frac{\sqrt{a^2t^2+1}-1}{a}$ is unique to the EOM associated with $\mathcal{L}$

Status Continued

- Reviewing dark matter problem
- Reviewing orbital motion $\mathcal{L} = \mathcal{L}_{CM} + \mathcal{L}_{rel}$

End